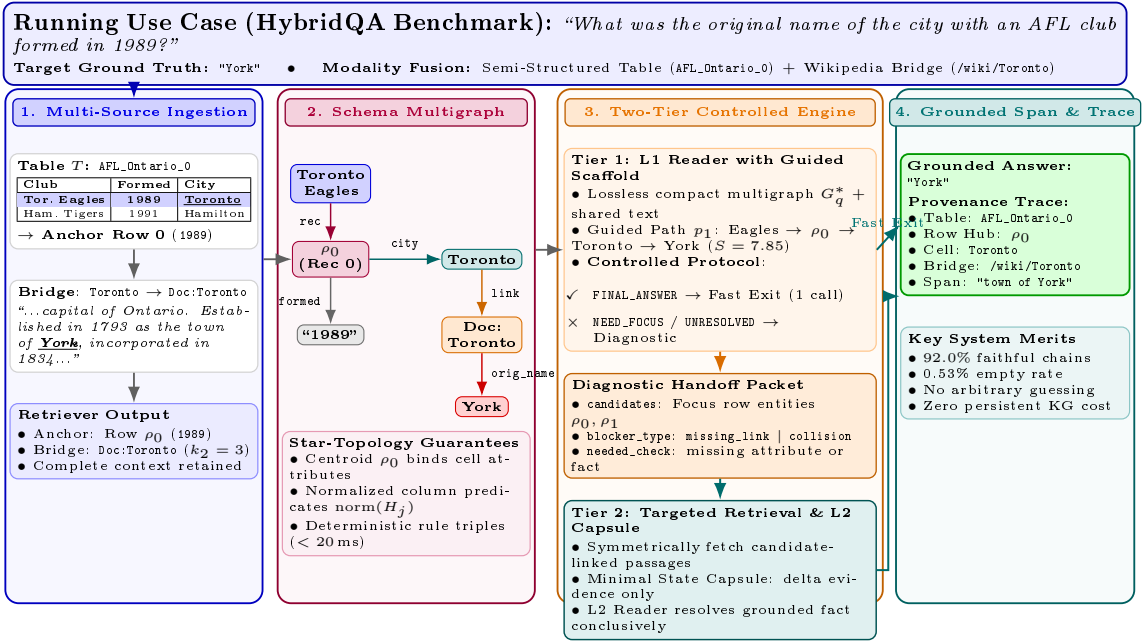

Running Use Case (HybridQA Benchmark): "What was the original name of the city with an AFL club formed in 1989?"
Target Ground Truth: "York" • Modality Fusion: Semi-Structured Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)
1. Multi-Source Ingestion
Table T: AFL_Ontario_0
Club | Formed | City
Tor. Eagles | 1989 | Toronto
Ham. Tigers | 1991 | Hamilton
→ Anchor Row 0 (1989)
Bridge: Toronto → Doc:Toronto
"...capital of Ontario. Established in 1793 as the town of York, incorporated in 1834..."
Retriever Output
• Anchor: Row ρ_0 (1989)
• Bridge: Doc:Toronto ($k_2 = 3$)
• Complete context retained
2. Schema Multigraph
Toronto Eagles
rec
ρ_0 (Rec 0)
city
Toronto
formed
"1989"
link
Doc: Toronto
orig_name
York
Star-Topology Guarantees
• Centroid ρ_0 binds cell attributes
• Normalized column predicates norm(H_j)
• Deterministic rule triples (< 20 ms)
3. Two-Tier Controlled Engine
Tier 1: L1 Reader with Guided Scaffold
• Lossless compact multigraph G_q^* + shared text
• Guided Path p_1: Eagles → ρ_0 → Toronto → York ($S = 7.85$)
• Controlled Protocol:
✓ FINAL_ANSWER → Fast Exit (1 call)
× NEED_FOCUS / UNRESOLVED → Diagnostic
Fast Exit
Diagnostic Handoff Packet
• candidates: Focus row entities ρ_0, ρ_1
• blocker_type: missing_link | collision
• needed_check: missing attribute or fact
Tier 2: Targeted Retrieval & L2 Capsule
• Symmetrically fetch candidate-linked passages
• Minimal State Capsule: delta evidence only
• L2 Reader resolves grounded fact conclusively
4. Grounded Span & Trace
Grounded Answer:
"York"
Provenance Trace:
• Table: AFL_Ontario_0
• Row Hub: ρ_0
• Cell: Toronto
• Bridge: /wiki/Toronto
• Span: "town of York"
Key System Merits
• 92.0% faithful chains
• 0.53% empty rate
• No arbitrary guessing
• Zero persistent KG cost